

Activity 1 – Initial Impressions

1. What insights are the researchers hoping to gain by collecting this data? What research questions are they hoping to answer?

The researchers aim to evaluate the effectiveness of the LinguaLift software by comparing users' learning outcomes before and after usage, and to examine how user satisfaction and engagement relate to performance improvements.

2. Participant 13's data differs from the others in the data set.

- a) What do you think could have happened and how might this affect the overall data analysis?

Participant 13 shows high usage time but very low satisfaction scores, suggesting a mismatch between engagement and user experience, possibly due to content difficulty or usability issues. This inconsistency could bias overall results and weaken the validity of conclusions.

- b) What strategies could you use to handle outliers in this data set?

Outliers can be handled by removing them, analyzing them separately, or applying robust statistical methods to minimize their impact on the overall results.

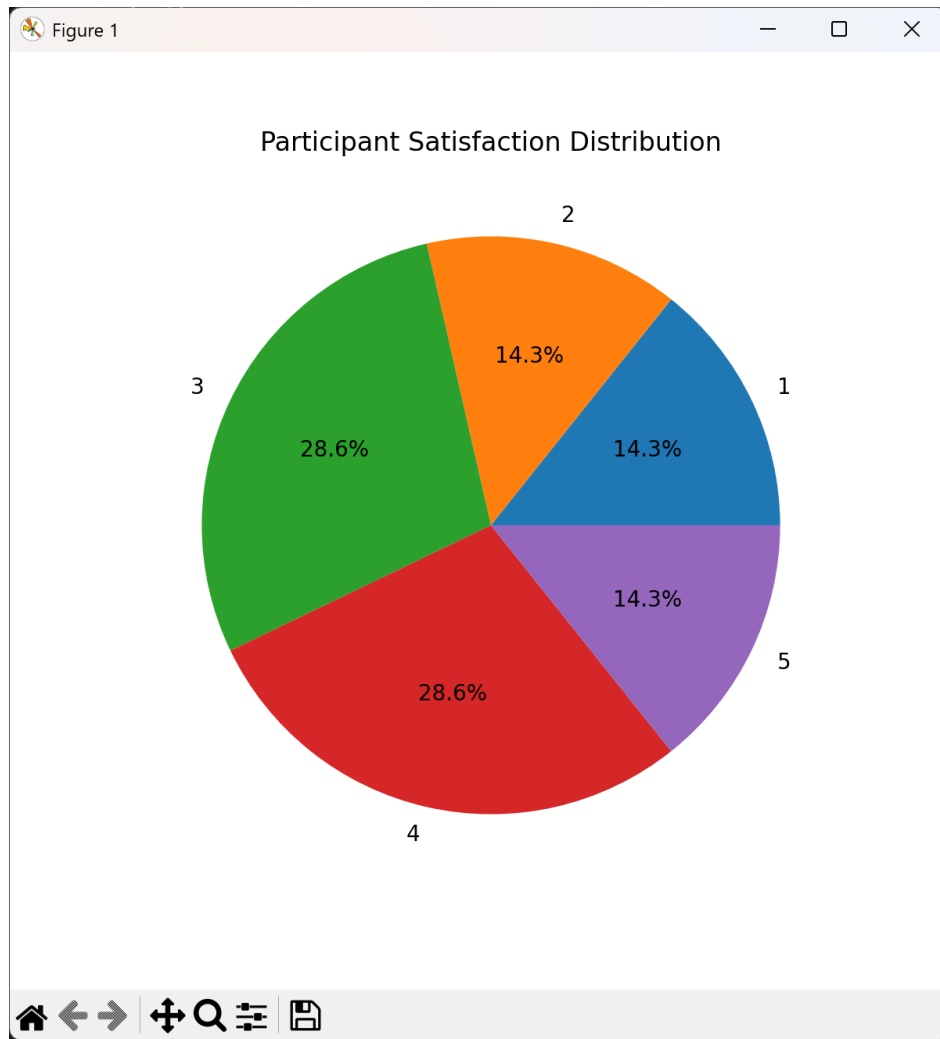
Activity 2 – Quantitative Data

3. What are some simple ways researchers might summarize the survey results to present to a team of software developers?

Researchers can summarize survey results using descriptive statistics such as mean scores and frequency distributions, as well as visualizations like bar charts or histograms to clearly present patterns to developers.

4. Using Datawrapper (<https://www.datawrapper.de/>), Excel, Python (matplotlib.pyplot) or any other software you are familiar with, create a plot that shows

- a) Participant satisfaction with the app



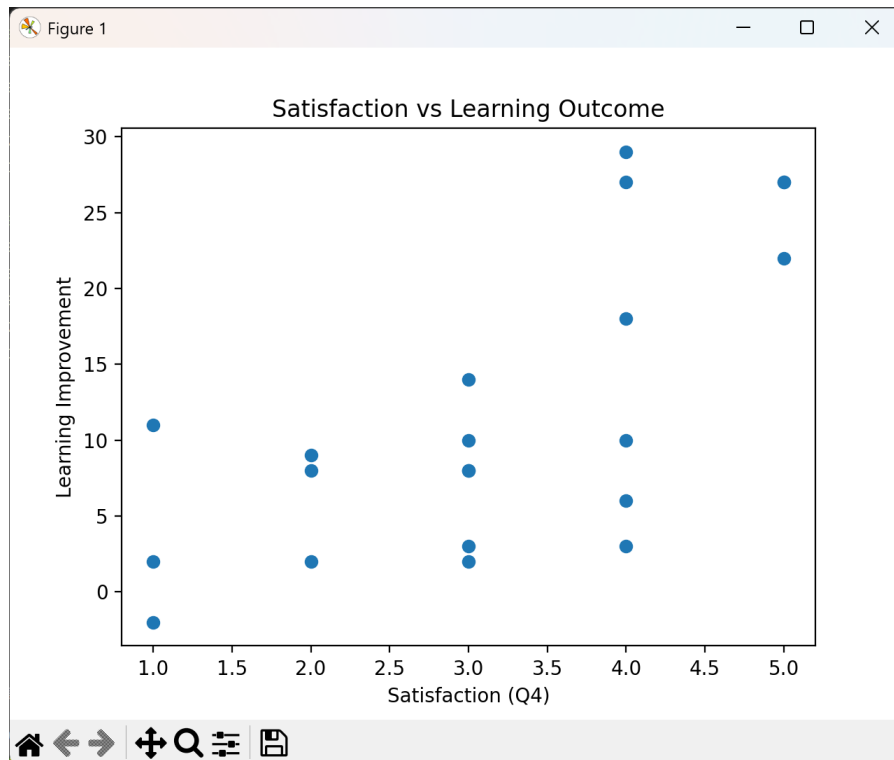
Justification:

A pie chart is used to show the proportion of participants in each satisfaction category.

Trend:

Most participants fall into the middle to high satisfaction levels (scores 3–4), indicating generally positive user satisfaction.

b) How participant satisfaction is related to learning outcomes



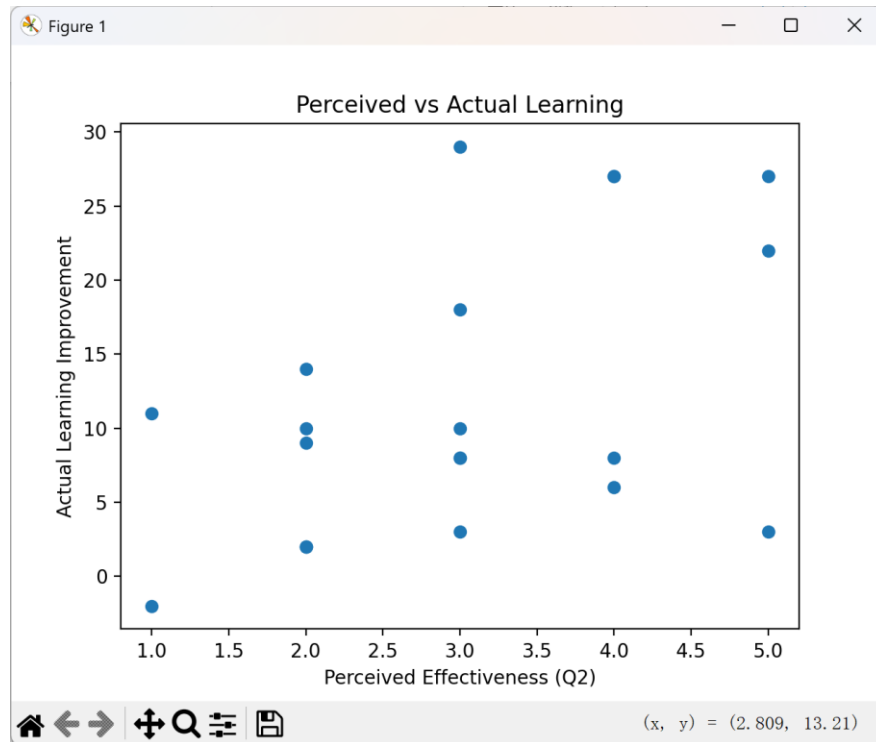
Justification:

A scatter plot is appropriate because it shows the relationship between satisfaction and learning improvement for each participant.

Trend:

There is a weak positive relationship, where higher satisfaction is generally associated with greater learning improvement.

c) The relationship between actual learning outcome and perceived learning outcomes of the participants



Justification:

A scatter plot is used to compare perceived learning effectiveness with actual learning outcomes.

Trend:

There is a general positive pattern, but some inconsistencies indicate that perceived effectiveness does not always match actual performance improvement.

Activity 3 – Qualitative Data

5. The research team has identified the following codes from the qualitative data:

Q5 (Most Useful Feature)	Code	Q6 (Biggest Issue)	Code
Interactive quiz game	Engagement Features	Occasional crashes during lessons	Technical Issues
AI conversations	AI Interaction	Complicated to get around	Usability Issues
Personalized learning paths	Learning Effectiveness	Needs more language options	Content Limitations and Diversity
Daily progress tracker	Learning Effectiveness	User interface is not intuitive	Usability Issues
Gamification	Engagement	Too few examples in	Content Limitations and

elements	Features	foreign language	Diversity
Quiz game and feedback	Engagement Features	Slow to load new lessons	Technical Issues
Multiple language support	Content Limitations and Diversity	Speech recognition needs improvement	AI Interaction
AI-driven practice conversations	AI Interaction	Lacks advanced content for proficient users	Content Limitations and Diversity
Variety of exercises	Engagement Features	Repetitive tasks	Content Limitations and Diversity
AI pronunciation feedback	AI Interaction	Inaccurate translations sometimes	AI Interaction
Progress tracking and learning path	Learning Effectiveness	Not enough cultural context	Content Limitations and Diversity
Custom learning pathway	Learning Effectiveness	Nothing, it's great	Engagement Features
The lessons are really engaging and fun	Engagement Features	Not much, a few games were a bit too easy	Content Limitations and Diversity
Interactive games and quizzes	Engagement Features	Limited number of stories	Content Limitations and Diversity
Grammar tips from AI	Learning Effectiveness	Some lessons are too long	Content Limitations and Diversity
Real-time feedback	Learning Effectiveness	Occasional technical glitches	Technical Issues
Cultural insights	Learning Effectiveness	Frequent crashes were annoying	Technical Issues
Learning games	Engagement Features	Some games were too easy	Content Limitations and Diversity
Extensive vocabulary	Learning	More examples needed in some	Content Limitations and

lists	Effectiveness	sections	Diversity
Flashcard game	Engagement Features	Would have loved more flashcard variety	Content Limitations and Diversity
Real-time feedback on pronunciation	Learning Effectiveness	A lot of the time AI didn't understand me	AI Interaction

Each participant comment was coded by identifying the main issue or feature being described and matching it to the most relevant category. For example, comments about navigation and interface were classified as usability issues, system performance problems as technical issues, and comments related to AI features were grouped under AI interaction. This approach ensures that each comment is categorized based on its primary focus.

6. One theme that could be developed from the qualitative data is “User Experience”. The User Experience theme encompasses both the issues that hinder experience as well as features that enhance it, including codes such as Usability Issues, Technical Issues, AI Interaction and Engagement Features.

Theme 1: Learning Quality

Another theme is “Learning Quality,” which focuses on how effectively the software supports language learning. This theme includes codes such as Learning Effectiveness and Content Limitations and Diversity, as many participants mentioned feedback, learning pathways, and the need for more advanced or diverse content.

Theme 2: System Reliability

A second theme is “System Reliability,” which relates to the stability and performance of the system. This theme includes Technical Issues and Usability Issues, as participants frequently reported crashes, slow performance, and difficulties navigating the interface.

These themes suggest that both learning support and system performance are key factors influencing overall user satisfaction and effectiveness.

Activity 4 – Improving the Study Design

7. Your group is tasked with re-designing the data collection methodology for the next round of testing. Discuss any changes you would make to the data collected through the software or in the feedback survey to improve the comprehensiveness and accuracy of the study.

To improve the study design, more detailed data could be collected, such as participant background information (e.g., prior language proficiency and learning goals) to better interpret learning outcomes. In addition, the survey could include more specific questions about different features (e.g., AI interaction, content difficulty) and use a larger sample size or longer testing period to improve the reliability and comprehensiveness of the results.

8. As a class, discuss the benefit of the mixed-method data set in this example. What information was gained from the mixed-method data that couldn’t be obtained from

quantitative or qualitative data alone?

The mixed-method data provides both quantitative and qualitative insights, allowing researchers to understand not only what the outcomes are, but also why they occur.

Quantitative data shows measurable improvements in learning performance, while qualitative data explains user experiences, such as usability issues and content limitations. This combination provides a more complete understanding than either method alone.